CURVED DUNN ADDITIVITY AT HIGHER GENUS: THE BRIDGE BETWEEN MODULAR-BOOTSTRAP AND TWISTING-COCHAIN COMPLEXES

RAEEZ LORGAT

Abstract. Classical Dunn additivity decomposes $E_2 \simeq E_1 \otimes E_1$. At genus $g \geq 1$, the modular bar complex of the curved modular A_∞-chiral operad is curved by $d^2_{\mathrm{fib}} = \kappa_{\mathrm{mod}}(A) \cdot \omega_g$ (where ω_g is the Arakelov form on $\overline{\mathcal{M}}_{g,n}$ and $\kappa_{\mathrm{mod}}(A)$ is the modular curvature coefficient of the chiral algebra); the Dunn decomposition is obstructed by R-matrix monodromy at sewing circles.

The core obstruction lives in H^2 of two different complexes: a modular-bootstrap complex and a curved-Dunn twisting-cochain complex. A degree-two comparison map Φ, when supplied, compares their obstruction classes. Genus 1 is a separate curved comparison governed by the Hodge bundle and the Faltings–Arakelov normalisation $\omega_1 = 12c_1(\lambda_1)$. The higher-genus vanishing statement starts at $g \geq 2$ and is conditional on modular-bootstrap H^2-acyclicity together with the degree-two comparison. KZB boundary composition is conditional on compatible boundary formal types and mixed Stokes data; a simple node is regular singular of Newton slope 0.

1. Genus-stratified obstruction

1.0.0.1. Genus 0 (uncurved Dunn). $\mathcal{O}^{A_\infty\text{-ch}}\big|_{g=0} = \mathrm{SC}^{\mathrm{ch,top}} \times E_1^{\mathrm{tr}}$. The bar complex factors as an ordinary tensor product.

1.0.0.2. Obstruction theory. The obstruction to extending the genus-0 product to genus g lies in $H^2(\mathrm{TCo}^\bullet_g(A), d_0)$, the twisting-cochain complex of §2. At genus 1: controlled by $\kappa_{\mathrm{mod}}(A) \cdot \omega_1$ via the monodromy $\mathrm{Mon}(R)$ of the classical r-matrix around the A-cycle. At genus $g \geq 2$: conditional on modular-bootstrap H^2-acyclicity and the degree-two comparison map Φ.

1.0.0.3. Genus-one twisted tensor product.

Proposition 1.1 (Genus-1 curved tensor-product comparison)**.** *Let A be a chirally Koszul algebra on a smooth projective curve X with nonzero modular curvature coefficient $\kappa_{\mathrm{mod}}(A)$. Assume that the A-cycle monodromy of the classical r-matrix defines a twisting cochain τ_1 in the genus-one Eilenberg–Zilber complex and that $d\tau_1 = \kappa_{\mathrm{mod}}(A)\omega_1$ as a curvature representative in the chosen coderived ambient category. Then*

$$\mathbf{B}_{\mathrm{mod}}\big(\mathcal{O}^{A_\infty\text{-}ch}\big)\big|_{g=1} \simeq \mathbf{B}_{\mathrm{mod}}\big(\mathrm{SC}^{\mathrm{ch,top}}\big)\big|_{g=1} \otimes^{\mathrm{Mon}(R)} \mathbf{B}\big(E_1^{\mathrm{tr}}\big),$$

where $\otimes^{\mathrm{Mon}(R)}$ is the Eilenberg–Zilber twisted tensor product with twisting cochain $\tau_1 = \mathrm{Mon}(R) = \exp\big(\oint_{A\text{-}cyc} r(z)\,dz\big)$.

2. Two complexes, one obstruction

Fix a chirally Koszul A on X at non-critical level, modular characteristic $\kappa_{\mathrm{mod}}(A) \neq 0$, and Arakelov form ω_g. At genus 1 we use the active normalisation $\omega_1 = 12c_1(\lambda_1)$.

Date: April 2026.

Definition 2.1 (Modular-bootstrap complex)**.** $(\mathfrak{g}^A_{\mathrm{mod}},\ d_{\Theta^{(0)}})$, the dg Lie algebra whose underlying graded vector space is $\prod_{g\geq 0,\, n\geq 0}\mathrm{Hom}_{\Sigma_n}(\mathcal{M}\mathrm{Ass}(g,n),\ \mathrm{End}_A(n))^{\Sigma_n}$, with differential $d_{\Theta^{(0)}} = [\Theta^{(0)}, -]$ the bracket with the genus-0 Maurer–Cartan seed. The genus tower $\{\Theta^{(g)}\}$ satisfies $D\Theta + \frac{1}{2}[\Theta,\Theta] = 0$ genus-wise. The degree-two vanishing used below is the modular-bootstrap input; the comparison with the curved-Dunn complex is a separate low-degree bridge condition.

Definition 2.2 (Curved-Dunn twisting-cochain complex)**.** $(\mathrm{TCo}^\bullet_g(A),\ d_0)$ at genus g with

$$\mathrm{TCo}^\bullet_g(A) \ = \ \mathrm{Hom}\big(\mathbf{B}(E_1^{\mathrm{tr}}),\ \mathrm{gr}^g\,\mathbf{B}_{\mathrm{mod}}(\mathrm{SC}^{\mathrm{ch,top}})\big),$$

differential $d_0(f) = \partial f - (-1)^{|f|} f\,\partial$ the convolution differential of the Eilenberg–Zilber twisted tensor product. The obstruction to extending $\tau_{<g}$ to τ_g is $\mathrm{Ob}_g \in H^2(\mathrm{TCo}^\bullet_g(A), d_0)$.

Remark 2.3 (Two-complex distinction)*.* Both complexes host the obstruction Ob_g but via different differentials on *a priori* different graded objects: bootstrap uses the tree-twisted Feynman transform of $\mathcal{M}\mathrm{Ass}$, Dunn uses the Eilenberg–Zilber machine on the tensor factorisation. The modular-bootstrap H^2 vanishing is an independent input. The curved-Dunn H^2 has no direct argument from the Arakelov class, since its differential is governed by the transverse E_1 factor. The bridge is the required comparison.

3. The bridge proposition

Proposition 3.1 (Modular-bootstrap to curved-Dunn comparison criterion)**.** *Assume there is a complete filtered L_∞-morphism*

$$\Phi \ : \ \big(\mathfrak{g}^A_{\mathrm{mod}},\ d_{\Theta^{(0)}}\big) \ \longrightarrow\ \big(\mathrm{TCo}^\bullet_g(A),\ d_0\big),$$

natural in A, such that:

(i) *the linear term Φ_1 is strict on the associated graded and induces an isomorphism on completed degree-two cohomology;*
(ii) *for every $g \geq 1$, the full L_∞ transport sends the bootstrap obstruction class to the curved-Dunn obstruction class $\mathrm{Ob}_g \in H^2(\mathrm{TCo}^\bullet_g(A))$;*
(iii) *the inverse-limit comparison has vanishing $\varprojlim^1$ in degree 2, so completed cohomology is computed by the associated graded comparison.*

Under this hypothesis, $H^2(\mathfrak{g}^A_{\mathrm{mod}}, d_{\Theta^{(0)}}) = 0$ implies $H^2(\mathrm{TCo}^\bullet_g(A), d_0) = 0$.

Proof. Work in the completed filtered complexes. Hypotheses (i) and (iii) identify the completed degree-two cohomology of the bootstrap and curved-Dunn complexes through the associated graded linear term Φ_1. The higher Taylor components of Φ transport Maurer–Cartan obstruction classes, so hypothesis (ii) identifies the obstruction class, not a preferred representative. If

$$H^2(\mathfrak{g}^A_{\mathrm{mod}}, d_{\Theta^{(0)}}) = 0,$$

then every transported degree-two curved-Dunn class is exact. In particular the obstruction class Ob_g vanishes in completed cohomology. □

Remark 3.2 (Concrete bridge data)*.* Both complexes are governed by the same underlying datum: R-matrix monodromy at sewing circles. Modular-bootstrap organises this datum by stable ribbon graph (Feynman transform); curved-Dunn organises it by twisting cochain (Eilenberg–Zilber). The candidate for Φ sends a stable ribbon graph Γ of type (g,n) with internal edges $e_1,\ldots,e_k$ to

$$\big(\mathrm{Mon}(R)_{e_1}\otimes\cdots\otimes\mathrm{Mon}(R)_{e_k}\big)\cdot \mathrm{ev}_\Gamma.$$

The bridge hypothesis is exactly the assertion that this candidate is a filtered L_∞ comparison through degree two, identifies the obstruction classes, and induces an isomorphism on completed H^2. The graph map must carry the automorphism weights and orientation cocycle; in particular the theta graph has weight 1/12 and the bouquet graph has weight 1/8. A model-categorical route through Bellier-Millès–Drummond-Cole would have to prove the same degree-two comparison.

4. Genus-one curved comparison: Gauss–Manin and Arakelov

Proof of Proposition 1.1 under the stated twisting-cochain hypothesis. The hypothesis supplies the only primitive used below: the genus-one twisting cochain τ_1 with $d\tau_1 = \kappa_{\mathrm{mod}}(A)\omega_1$ in the selected coderived model. The proof verifies that this input gives the displayed twisted tensor product.

Step 1 (Gauss–Manin connection on $\Lambda^2 H^1$). The universal elliptic curve $\mathcal{E} \to \overline{\mathcal{M}}_{1,1}$ carries the Hodge bundle $\lambda_1 = \pi_*\Omega^1_{\mathcal{E}/\overline{\mathcal{M}}_{1,1}}$ and its second wedge $\lambda_2 = \Lambda^2 R^1\pi_*\mathbb{Q}$. The Gauss–Manin connection ∇^{GM} on $R^1\pi_*\mathbb{Q}$ descends to λ_2 as a flat connection on a rank-1 bundle. On the curve side, the bar-complex differential at genus 1 factorises through ∇^{GM} via the fibrewise Hochschild complex:

$$d_{\mathbf{B}_{\mathrm{mod}},\, g=1} \;=\; d_{\mathrm{fib}} \;+\; \kappa_{\mathrm{mod}}(A)\cdot \omega_1 \wedge (\cdot),$$

with $\omega_1 = 12c_1(\lambda_1)$ in the Faltings–Arakelov normalisation. The Arakelov class is a base class. The comparison is representative-level inside the chosen Eilenberg–Zilber complex. It does not assert that the Hodge or Arakelov class is exact on $\overline{\mathcal{M}}_{1,1}$.

Step 2 (Arakelov pairing on λ_1). The Faltings–Arakelov metric on the Hodge bundle fixes the representative of $\omega_1 = 12c_1(\lambda_1)$. We use this normalisation only as a curvature representative on the base moduli stack. The Arakelov pairing $\langle\cdot,\cdot\rangle_{\mathrm{Ar}}$ on λ_1 is ∇^{GM}-compatible, giving an isomorphism of flat bundles $\lambda_2 \otimes \lambda_1 \simeq \lambda_1^\vee$ up to an integer twist. Pulling back along the chosen genus-one comparison, the twisted Künneth factorisation becomes

$$\mathbf{B}^{(1)}_{\mathrm{mod}}(A) \;\simeq\; \mathbf{B}^{(1)}_{\mathrm{mod}}(\mathrm{SC}^{\mathrm{ch,top}}) \otimes \mathbf{B}(E_1^{\mathrm{tr}}) \quad \text{twisted by} \quad \mathrm{Mon}(R) \;=\; \exp\Big(\oint_{A\text{-cyc}} r(z)\,dz\Big).$$

The exponent is the A-cycle period of the classical r-matrix, a well-defined element of $\mathrm{End}_A \otimes \mathrm{End}_A$ because $r(z)$ satisfies the KZ integrability condition.

Step 3 (Eilenberg–Zilber compatibility). The Eilenberg–Zilber twisted tensor product with twisting cochain $\tau = \mathrm{Mon}(R)$ has differential $\partial_\tau(x\otimes y) = \partial x \otimes y + (-1)^{|x|} x \otimes \partial y + \tau\cdot(x\otimes y)$. The assumption $d\tau_1 = \kappa_{\mathrm{mod}}(A)\omega_1$ identifies this differential with the curved differential at genus 1. The factorisation is an isomorphism in the stated coderived ambient category. □

Remark 4.1 (Genus-one bridge). The genus-one bridge is the first nonzero test case for the curved-Dunn comparison: Gauss–Manin uncurving on $\Lambda^2 H^1$, the Faltings–Arakelov class, and the Eilenberg–Zilber twisting cochain must give the same degree-two obstruction class.

Remark 4.2 (Relationship to the bridge at genus 1). At $g = 1$, the bridge comparison Φ of Proposition 3.1 reduces to the assertion that the annular bootstrap insertion $\Theta^{(1)}$ and the curved-Dunn twisting cochain τ_1 represent the same monodromy class. At $g \geq 2$, the comparison is not tautological: higher-genus stable-graph strata must match higher Eilenberg–Zilber corrections.

5. Conditional higher-genus curved-Dunn H^2 vanishing

Theorem 5.1 (Conditional curved-Dunn H^2 vanishing for $g \geq 2$). *Assume the modular-bootstrap degree-two vanishing and the low-degree bridge hypothesis of Proposition 3.1. For every $g \geq 2$ and every chirally Koszul A in the standard landscape at non-critical level,*

$$H^2\big(\mathrm{TCo}^\bullet_g(A),\; d_0\big) \;=\; 0.$$

Consequently, $\tau_{<g}$ extends to τ_g for every $g \geq 2$ up to convolution-gauge equivalence, and the modular bar complex factorises cohomologically as a twisted Künneth tensor product:

$$\mathbf{B}_{\mathrm{mod}}\big(\mathcal{O}^{A_\infty\text{-}ch}\big) \;\simeq\; \mathbf{B}_{\mathrm{mod}}\big(\mathrm{SC}^{\mathrm{ch,top}}\big) \;\otimes^{\tau}\; \mathbf{B}\big(E_1^{\mathrm{tr}}\big).$$

Proof. $H^2(\mathfrak{g}^{A}_{\mathrm{mod}}, d_{\Theta^{(0)}}) = 0$ at every $g \geq 2$ by the modular-bootstrap input. The genus-one curved comparison of Proposition 1.1 does not enter as a vanishing base case. The Maurer–Cartan recursion exhibits the obstruction class; it does not by itself prove vanishing.

Apply Proposition 3.1(iii): the assumed isomorphism $H^2(\mathfrak{g}^{A}_{\mathrm{mod}}) \simeq H^2(\mathrm{TCo}^{\bullet}_{g}(A))$ transports the vanishing. The extension $\tau_{<g} \mapsto \tau_g$ is automatic: $\mathrm{Ob}_g = [d_0(\tau_{<g}) + \frac{1}{2}[\tau_{<g}, \tau_{<g}]_{\mathrm{conv}}] = 0$ in H^2, so the cocycle is a coboundary and the twisting cochain extends. □

6. Boundary KZB formal type and Stokes composition

At non-critical level, smooth-locus KZB flatness does not by itself define irregular modular-operad composition at the boundary. The needed input is a compatible meromorphic formal type on every stable-graph boundary stratum, together with mixed-channel Stokes data at corners. Edgewise Stokes data alone do not determine two-edge clutching.

Fix a point $p \in \partial\overline{\mathcal{M}}_{g,n}$ on the boundary nodal divisor, with local coordinate ${}_p$ transverse to the divisor. The KZB connection near p has the form $\nabla^{\mathrm{KZB}} = d + \Omega^{\mathrm{KZB}}({}_p)$. At a simple node the polar part is logarithmic: $\Omega^{\mathrm{KZB}}({}_p) = A_1\, d{}_p / {}_p + \Omega_{\mathrm{reg}}$. The singularity is regular, with Newton slope 0; it is not a Stokes source. Genuine irregular Stokes data enter only on strata whose formal boundary model contains higher-order terms after ramification.

Theorem 6.1 (Stokes-sector composition from boundary KZB formal type). *Assume the boundary formal-type theorem: on every stable-graph boundary stratum, including two-edge corners, the KZB connection has compatible formal types, regular-singular slope-0 simple-node restrictions, residue-gap nonresonance in the irregular sectors, and vanishing mixed-channel Stokes curvature. Then for every (g, n) at non-critical level, the resulting formal/Stokes boundary data define a factorisation map on the modular operad $M\,\mathrm{Ass}(g, n)$. Consequently, modular-operad composition in the irregular boundary-formal-type regime is defined for all (g, n).*

Proof under the boundary formal-type hypothesis. At a simple node the logarithmic residue supplies regular-singular monodromy and no Stokes filtration. On every boundary component with nonzero irregular slope, the assumed formal type gives the exponential factors, Stokes directions, and Stokes groups after finite ramification. Jimbo–Miwa isomonodromic deformation and the formal/Stokes classification of meromorphic connections identify the sectorial continuation maps of flat sections; the Stokes matrices are additional monodromy data constrained by the formal type and by the isomonodromy equations.

These one-edge operations do not imply coherence at a two-edge corner. The assumed mixed-channel zero-curvature condition is exactly the statement that the two orders of clutching yield the same sectorial continuation. With this corner compatibility imposed on every stable-graph stratum, Getzler–Kapranov clutching composes the regular-singular simple-node monodromies and the nonzero-slope Stokes sector maps into a well-defined modular-operad composition. □

Corollary 6.2 (Modular operad composition under boundary formal type). *Under the boundary formal-type theorem, the modular A_∞-chiral operad $\mathcal{O}^{A_\infty\text{-}ch}$ admits operadic composition at integer levels (classical regular-singular KZB) and at non-critical levels in the supplied formal-type sectors (regular-singular simple-node monodromy plus Stokes-sector composition on nonzero-slope boundary strata), uniformly at all (g, n) satisfying the hypothesis.*

7. Summary

(1) **Bridge.** Proposition 3.1 states the degree-two comparison criterion between the modular-bootstrap and curved-Dunn obstruction complexes.
(2) **Genus-1 curved comparison.** Proposition 1.1 uses Gauss–Manin on $\Lambda^2 H^1$ and Arakelov pairing on λ_1 under the stated twisting-cochain hypothesis.
(3) **Higher-genus vanishing.** Theorem 5.1: curved-Dunn $H^2 = 0$ for $g \geq 2$ under modular-bootstrap acyclicity and the degree-two bridge hypothesis; the twisting cochain extends up to gauge and the twisted tensor factorisation holds cohomologically.
(4) **Boundary KZB formal type.** Theorem 6.1: modular-operad composition in the irregular boundary-formal-type sector under compatible formal types, residue-gap nonresonance, and mixed Stokes data.

Under these hypotheses, the curved-Dunn decomposition has the cohomological, gauge-qualified form stated above: the modular bar complex is modeled by a twisted Eilenberg–Zilber tensor product of the closed-colour $\mathrm{SC}^{\mathrm{ch,top}}$ modular bar with the open-colour transverse E_1 bar, with twisting cochain built recursively from R-matrix monodromy at genus-raising clutching.

References

[1] R. Lorgat, *Chiral bar and chiral cobar: Koszul duality, modular invariants, and the quantum geometry of vertex algebras*, Volume I (2026).
[2] R. Lorgat, *A_∞-chiral algebras and 3D holomorphic-topological QFT*, Volume II (2026).
[3] M. Jimbo and T. Miwa, *Monodromy preserving deformation of linear ordinary differential equations with rational coefficients. II*, Physica D 2 (1981).
[4] J. Bellier-Millès and G. Drummond-Cole, *Model structures on curved operads* (preprint, 2024).